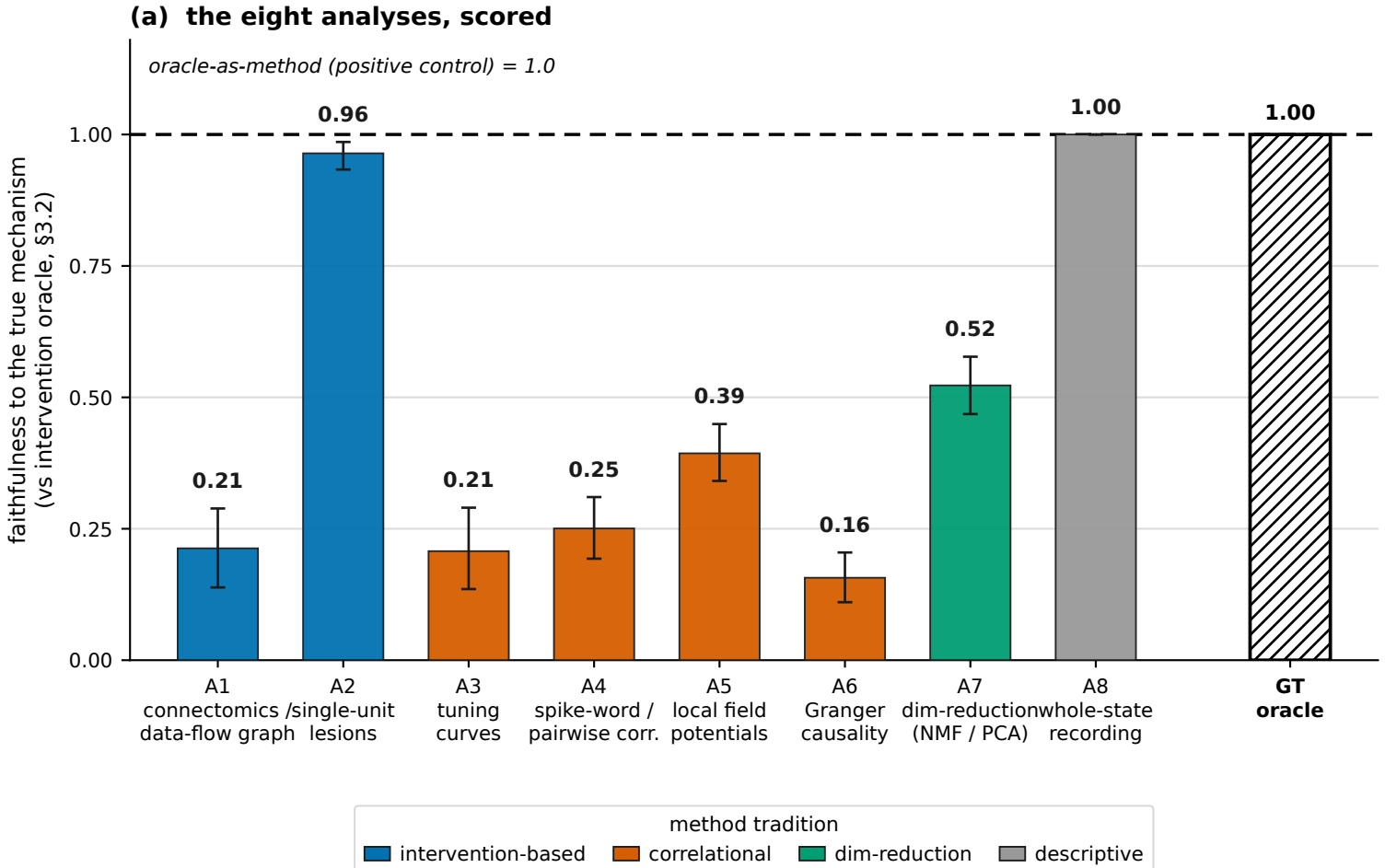
The neuroscience battery on a known machine — scored against ground truth

Jonas & Kording's battery, quantified: classical analyses find rich structure yet score low in faithfulness to the true mechanism.



(b) key findings

A6 **Granger F = 0**

Pong false-edge 1.00, missed 1.00 → F = 0.00

Temporal precedence ≠ causation: a clock-locked machine yields spurious bidirectional edges.

A3 **spurious tuning**

Pong rate 0.50; Space Inv. 0.33; Q*bert 0.53

Strongly-tuned units whose tuning ≠ true causal role — the tuning-curve trap, quantified.

A2 **high ρ, blind to interactions**

ρ = 0.96 vs true role; misses 15% of interactions

Super-additive pair Δ up to 40 (Space Inv.) is invisible to one-unit-at-a-time ablation.

A7 **NMF > PCA (both mediocre)**

NMF matched frac 0.52 vs PCA 0.34 (best 1.0)

Neither recovers the true basis; both score well below the oracle.

Data: committed §R records (44-game scored battery); whiskers = 95% bootstrap CI (every bar has a CI record). See Supplement provenance table.